

# Unified Lifelong Action Learning: A Real-Video Pipeline for Continual Learning, Knowledge Distillation, and Few-Shot Meta-Adaptation on UCF101

First A. Author  
Dept. of Computer Science  
University Name  
author1@example.com

Second B. Author  
Dept. of Computer Science  
University Name  
author2@example.com

Third C. Author  
Dept. of Computer Science  
University Name  
author3@example.com

**Abstract**—We present *Unified Lifelong Action Learning* (ULAL), an end-to-end pipeline that studies how a single video action recogniser can keep learning new actions over time without forgetting the old ones, trained directly on real video frames with no cached features. The pipeline is built as five linked studies, each answering one question. (i) *How do we measure honestly?* We show that UCF101’s shipped split leaks 80% of its source groups across train and test and replace it with a group-disjoint split that leaks none. (ii) *Which backbone?* A five-way bake-off selects a ResNet50+LSTM teacher (93.9% on the base classes). (iii) *How do we stop forgetting?* Across a  $10 \rightarrow 15 \rightarrow 20$  class stream, rehearsal reaches 89.8% while plain fine-tuning collapses to 26.0% and every data-free regulariser (EWC, LwF) trails. (iv) *How do we shrink the model?* Knowledge distillation compresses the teacher into a MobileNetV3-Small student that keeps 82.7% at roughly  $14\times$  fewer parameters, with cosine feature distillation best. (v) *How do we learn brand-new classes from a handful of clips?* A nine-arm meta-learning comparison, controlled by a matched fine-tune baseline that sees the same auxiliary data, shows *MAML+replay* is the best-balanced few-shot adapter (harmonic mean 0.60) and beats the fine-tune control by 0.07, isolating the meta-learning effect. For every stage we first explain *why* the methods should rank as they do from the ability each one has, then show the measured result. A final capstone notebook chains the winning choice of every stage and is described here pending its run. The whole pipeline runs on a single free-tier Colab T4 GPU.

**Index Terms**—continual learning, catastrophic forgetting, action recognition, knowledge distillation, meta-learning, few-shot learning, model compression.

## I. INTRODUCTION

Deep video models forget. Fine-tuning a trained action recogniser on a set of new classes collapses its accuracy on the classes it already knew—*catastrophic forgetting* [1]. Lifelong, or continual, learning aims to add classes over time while preserving what was already learned [5]. Video makes this markedly harder than images: a clip is a large spatio-temporal tensor, so storing and replaying past data is expensive, and the temporal model on top of the per-frame features adds a second place for representations to drift.

A realistic deployment does not need only one of these abilities; it needs several at once. It must (a) be measured on

data it has never seen, (b) start from strong features, (c) retain old classes as new ones arrive, (d) be small enough to run on a modest device, and (e) still pick up a brand-new class from only a handful of examples. Each of these is usually studied in isolation. The contribution of this paper is to study them *together*, as one pipeline on one dataset, trained *directly on real video frames* so that every reported number reflects the model that would actually be deployed—no offline feature cache stands between the study and the recogniser.

We build the pipeline on UCF101 [12] as five linked studies, each a single notebook answering one question, and each feeding the next. The organising idea is a *moving freeze boundary*: the recogniser is one shared network (a CNN backbone applied per frame, an LSTM over the frame features, and a classifier head that *grows* as classes arrive), and each stage decides how much of that network stays trainable. Early stages train broadly; later stages freeze the lower backbone to protect old features and let only the upper blocks, the LSTM, and the head move. This single design connects all five studies and lets each comparison change exactly one mechanism at a time.

### Contributions.

- **An honest measurement protocol.** We quantify the 80% source-group leakage in UCF101’s shipped split and replace it with a group-disjoint split that leaks nothing, so every later number is trustworthy though lower.
- **A controlled, five-stage study** of backbone selection, five continual-learning methods, four distillation objectives, and a nine-arm meta-learning comparison, all sharing one recogniser and one hyperparameter budget per stage.
- **A fine-tune control that isolates meta-learning.** In the few-shot stage the MAML init is compared against an ordinary fine-tune init that saw the *same* auxiliary data, so any remaining gain is attributable to meta-learning and not to extra data.
- **A balanced metric for the few-shot regime.** Because base classes outnumber new ones, we report the harmonic mean of base and new accuracy, which reveals a ranking

that a base-dominated average hides.

- **Explanations, not just tables.** For every stage we state *why* one method should beat another from its ability before showing the measured result, and we close with a cross-stage synthesis of when each tool wins.

## II. RELATED WORK

**Continual learning.** Approaches fall into three families. *Regularisation* methods add a penalty that discourages changing parameters important to old tasks: EWC [1] weights the penalty by the Fisher information, and synaptic-intelligence variants use online importance estimates [2]. *Functional* methods keep the model’s outputs stable: Learning without Forgetting (LwF) [3] distills the old model’s own predictions on new data. *Rehearsal* methods store and replay a small set of past exemplars; iCaRL [4] combines rehearsal with distillation and a nearest-mean classifier. We evaluate a representative of each family on video, where the cost of storing exemplars is a first-order concern. **Knowledge distillation.** A compact student can be trained to match a larger teacher’s softened logits [6] or its intermediate features [7]; we compare logit distillation against two feature objectives (cosine and a regressor-MSE) following the structure of the PyTorch distillation tutorial. **Meta and few-shot learning.** MAML [8] learns an initialisation from which a few gradient steps adapt to a new task; first-order MAML drops the second-order term for efficiency. Few-shot class-incremental learning (FSCIL) [9] adds new classes from few examples while retaining a large base set, and it is standard there to report the harmonic mean of base and new accuracy so that neither side can be ignored. Our few-shot stage sits in this setting and adopts that metric.

## III. SYSTEM, NOTATION, AND PROTOCOL

### A. The shared recogniser

Every stage uses the same recogniser, sketched in Fig. 2. A clip is  $T=16$  RGB frames at  $224 \times 224$ , ImageNet-normalised. A CNN backbone  $\phi$  is applied to each frame independently, producing a sequence of frame features  $\phi(x_1), \dots, \phi(x_T)$ . An LSTM reads that sequence and its final hidden state  $h_T \in \mathbb{R}^{256}$  is the clip embedding. A linear head  $W$  maps  $h_T$  to class logits. Formally, for a clip  $x$ ,

$$h_T = \text{LSTM}(\phi(x_1), \dots, \phi(x_T)), \quad z = W h_T. \quad (1)$$

The head *grows* as classes arrive: adding  $k$  classes appends  $k$  rows to  $W$  and keeps the existing rows, so old-class logits are never discarded, only joined by new ones. We call the shared clip embedding  $h_T = \varphi(x)$  the model’s *feature*; it is what distillation aligns and what the prototype probe reads.

### B. The moving freeze boundary

The five stages differ in *which* parameters they train, and this is the thread that ties the pipeline together. Stage 2 trains the whole recogniser to obtain strong base features. Stage 3 keeps most of the backbone plastic but only the last block is fine-tuned as classes are added, so that low-level features stay put. Stage 4 trains a separate small student end-to-end to match the

teacher. Stage 5 freezes the lower backbone and meta-trains only the upper blocks, the LSTM, and the head, so the old-class features are preserved while the upper layers reshape for the new task. Making this boundary explicit is what lets each comparison be fair: within a stage every method trains the *same* parameters, so only the mechanism under test differs.

### C. Datasets and budget

All experiments use UCF101 on a single free-tier Colab T4 GPU. The group-disjoint split assigns whole source groups to a single split in an 18/3/4 groups-per-class ratio (train/val/test), preserving the standard  $\approx 72/12/16\%$  proportions while removing leakage (Section IV). Table I lists the per-stage hyperparameters; every method compared within a stage shares them, and seeds are fixed and reset before each training run, so a difference in the result comes only from the mechanism under test.

TABLE I  
MAIN HYPERPARAMETERS PER STAGE (SHARED ACROSS THE METHODS COMPARED IN THAT STAGE). THIS IS THE “BUDGET” THAT MAKES EACH COMPARISON FAIR.

| Stage             | Key settings                                                                                       |
|-------------------|----------------------------------------------------------------------------------------------------|
| Backbone (S2)     | Adam, lr $10^{-4}$ , 5 epochs                                                                      |
| Continual (S3)    | Adam, lr $10^{-4}$ , 5 epochs/task; replay 15/class;<br>LwF weight 3.0, $T=5$ ; EWC $\lambda=2800$ |
| Distillation (S4) | Adam, lr $10^{-4}$ , 15 epochs; $T=2$ , CE/KD 0.75/0.25                                            |
| Meta-train (S5)   | 5-way 3-shot, 60 episodes $\times$ 8 epochs,<br>inner SGD 5 steps @ 0.05, meta-lr $10^{-4}$        |
| Adaptation (S5)   | 8 clips/class, 5 epochs, lr $3 \times 10^{-4}$ ;<br>replay 8/class; LwF 0.5, $T=2$                 |

### D. How to read the numbers

*Test accuracy* is the fraction correct on the unseen, group-disjoint test set. *Old* and *New* accuracy split that score into classes learned earlier versus the newest task, and *Base* is retention on the original classes. In the few-shot stage, where 20 base classes outnumber 5 new ones, a plain average is 20/25 dominated by the base and rewards a method that simply guards old classes while learning nothing new. We therefore add the *balanced* mean  $\frac{1}{2}(\text{acc}_{\text{new}} + \text{acc}_{\text{base}})$  and the *harmonic* mean

$$H = \frac{2 \text{acc}_{\text{new}} \text{acc}_{\text{base}}}{\text{acc}_{\text{new}} + \text{acc}_{\text{base}}}, \quad (2)$$

which weight the two sides equally and, for the harmonic, collapse to zero if either side is sacrificed. This is the standard FSCIL metric [9].

## IV. STAGE 1 — DATA SPLIT AND LEAKAGE

**Why this comes first.** A UCF101 filename `v_Class_gXX_cYY.avi` encodes a *group* gXX (one source recording) and a *clip* cYY (one segment of that recording). All clips of a group share nearly identical frames—the same actor, background, and camera. If the split is made per clip, segments of the *same* recording land

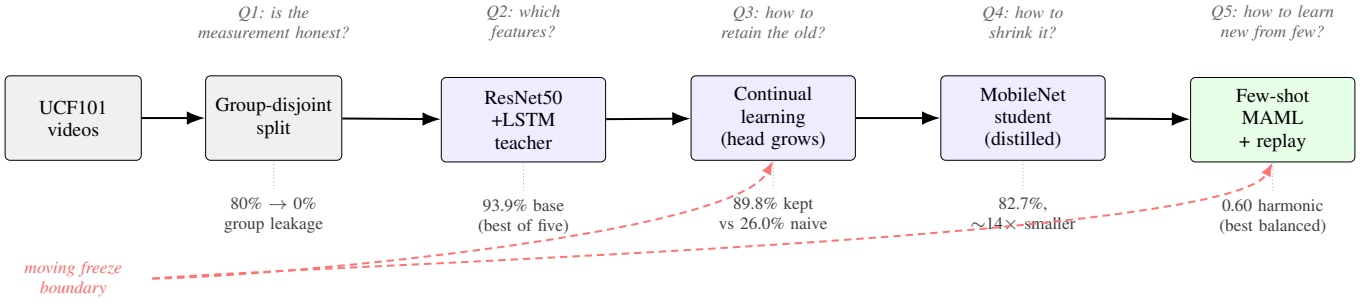


Fig. 1. **The ULAL pipeline.** Each stage answers one question (top), transforms the model (middle), and reports a headline result (bottom). UCF101 video is first split by source group so that no recording leaks across train and test (Stage 1). A ResNet50+LSTM teacher is trained on the base classes (Stage 2), extended to new classes by continual learning with a growing classifier head (Stage 3), compressed into a small MobileNet student by knowledge distillation (Stage 4), and finally adapted to brand-new classes from only a few clips with MAML plus replay (Stage 5). The dashed red arrows mark the *moving freeze boundary*: from Stage 3 onward the lower backbone is frozen to protect old features while the upper blocks, LSTM, and head remain trainable. Each stage supplies an ability the next one needs.

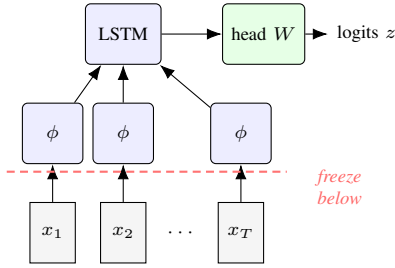


Fig. 2. **The shared recogniser.** The backbone  $\phi$  runs on each of the 16 frames; an LSTM summarises the sequence into a 256-d clip feature; a growing head classifies. The moving freeze boundary (dashed) decides how deep the trainable region reaches at each stage—broad early on, then narrowed to the upper backbone, LSTM, and head to protect old features.

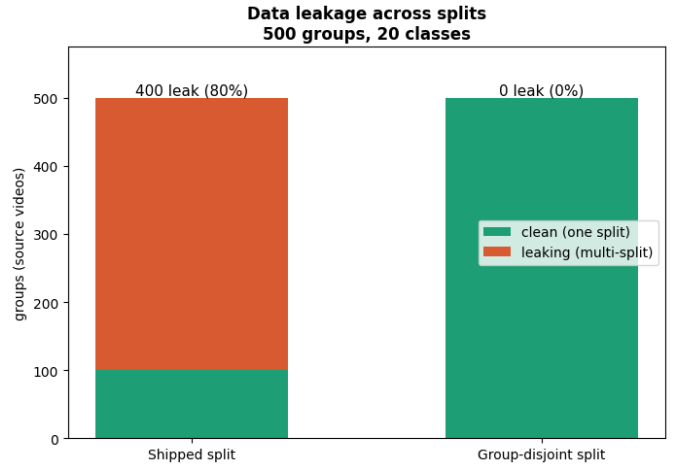


Fig. 3. Source groups that leak across the splits: 80% under the shipped per-clip split versus 0% under the group-disjoint split. Every subsequent number in this paper is read against the clean split; the accuracies are therefore lower than commonly reported on UCF101, but honest.

in both train and test, and the model is scored on footage it effectively trained on: it can recognise the *scene* rather than the *action* and appear far stronger than it is. No later comparison can be trusted until this is fixed, so we assign each whole group to a single split.

**Result.** Across the 20 studied classes (2,679 videos, 500 groups), the shipped split leaks **400 of 500 groups (80%)** across train/val/test, whereas the group-disjoint split leaks **0** (Fig. 3). The disjoint split keeps the conventional  $\approx 72/12/16\%$  proportions, so the drop in accuracy that follows is purely the removal of an inflation, not a change of task difficulty.

**Reading it.** The lesson is methodological rather than architectural: a dataset’s default split can silently reward memorising context. Because the same leakage would inflate every stage below—backbone, continual learning, distillation, and meta-adaptation alike—fixing it once at the source is what makes the rest of the study comparable to itself and to future work.

## V. STAGE 2 — BACKBONE SELECTION

**Why we expect differences.** The backbone supplies the per-frame features; the LSTM only has to combine them over time. A network with no prior knowledge (a from-scratch CNN) must learn good spatial features from a few thousand clips

and should struggle, whereas ImageNet-pretrained backbones already carry rich features and should cluster high. Among the pretrained backbones the choice becomes a trade-off of accuracy, feature dimensionality, and speed, so we compare five and select the best balance as the teacher for every later stage.

**Result.** Table II gives the measured numbers. The from-scratch CNN is far behind (52.8%); all four pretrained backbones exceed 90%. **ResNet50 is both the most accurate (93.9%) and a good size/speed balance**, so it becomes the teacher.

**Why the ordering.** The from-scratch CNN has no prior features *and* its LSTM ingests a large flattened feature map, giving it the most parameters (103M) yet the worst accuracy: capacity without useful priors does not help on little data. The pretrained backbones all reach the low-90s because the hard part—good spatial features—is already solved before the LSTM sees anything. ResNet50 edges the others thanks to

TABLE II  
BACKBONE BAKE-OFF ON THE 10 BASE CLASSES (GROUP-DISJOINT TEST). EVERY BACKBONE IS TRAINED UNDER THE IDENTICAL STAGE-2 BUDGET AND SEED.

| Backbone                  | Test acc. (%) | Params (M) | ms/clip |
|---------------------------|---------------|------------|---------|
| ScratchCNN                | 52.8          | 103.1      | 10.2    |
| ResNet18                  | 90.1          | 12.0       | 15.7    |
| DenseNet121               | 92.0          | 8.3        | 53.1    |
| VGG19-BN                  | 92.0          | 20.8       | 115.7   |
| <b>ResNet50 (teacher)</b> | <b>93.9</b>   | 25.9       | 54.6    |

its deeper, richer 2048-d feature, which also hands the later distillation and meta stages a clean feature space to build on; its 55 ms/clip is acceptable and less than half of VGG19-BN’s 116 ms. We therefore trade a slightly larger parameter count for the best accuracy and the best downstream feature.

## VI. STAGE 3 — CONTINUAL LEARNING

**The five methods and why they should differ.** Starting from the base teacher we grow the head and train on a  $10 \rightarrow 15 \rightarrow 20$  class stream with five methods. Their abilities predict the outcome. *Naive* fine-tuning minimises only the new-task loss,

$$\mathcal{L}_{\text{naive}} = \mathcal{L}_{\text{CE}}(\text{new}), \quad (3)$$

so it has no mechanism to protect the past and should forget badly. *EWC* adds a quadratic penalty that resists moving parameters important to old tasks,

$$\mathcal{L}_{\text{EWC}} = \mathcal{L}_{\text{CE}}(\text{new}) + \frac{\lambda}{2} \sum_i F_i (\theta_i - \theta_i^*)^2, \quad (4)$$

where  $F_i$  is the Fisher importance and  $\theta^*$  the old weights; it stores no data and can only *slow* drift. *LwF* keeps the old-class outputs stable by distilling the old model’s predictions on the new data,

$$\mathcal{L}_{\text{LwF}} = \mathcal{L}_{\text{CE}}(\text{new}) + \lambda \text{KD}(z_{1:C_{\text{old}}}^{\text{old}}, z_{1:C_{\text{old}}}^*), \quad (5)$$

a stronger signal than a weight penalty but still without ever seeing old data. *Replay* re-exposes the model to a few stored old clips  $\mathcal{R}$  mixed into each new batch  $\mathcal{B}$ ,

$$\mathcal{L}_{\text{replay}} = \mathcal{L}_{\text{CE}}(\mathcal{B} \cup \mathcal{R}), \quad (6)$$

directly refreshing the old decision boundaries. *Replay+LwF* [4] adds output stability on top of re-exposure. All arms share the schedule of Table I; only the mechanism differs.

**Result.** Table III reports accuracy after the *full* 20-class stream, split into old classes (0–14) and the newest task (15–19), plus base (0–9) retention. Figure 4 contrasts naive forgetting with replay across the whole stream.

**Why the winners win.** The table follows exactly from the abilities. *Naive* and *LwF* learn the newest task almost perfectly (94.9% and 100%) but erase everything older (old  $\leq 5\%$ ): with the shared weights free to move and no old data, the network re-purposes itself entirely for the last task. *LwF* even fits the new task *better* than naive because its

TABLE III  
CONTINUAL LEARNING AFTER THE FULL  $10 \rightarrow 15 \rightarrow 20$  STREAM (%). FINAL: ALL 20 CLASSES; OLD: CLASSES 0–14; NEW: LAST TASK 15–19; BASE: ORIGINAL 10. SORTED BY FINAL ACCURACY.

| Method        | Final       | Old (0–14) | New (15–19) | Base (0–9) |
|---------------|-------------|------------|-------------|------------|
| Naive         | 26.0        | 0.9        | 94.9        | 0.5        |
| LwF           | 30.5        | 5.3        | 100.0       | 0.0        |
| EWC           | 62.0        | 71.1       | 36.8        | 76.9       |
| Replay+LwF    | 87.0        | 82.6       | 99.2        | 83.0       |
| <b>Replay</b> | <b>89.8</b> | 86.7       | 98.3        | 89.2       |

distillation term stabilises optimisation, yet it still forgets—keeping *outputs* stable is no substitute for *seeing* old data. *EWC* is the opposite failure: its penalty protects the base (76.9%) but is so restrictive that the new task barely learns (36.8%), a stability/plasticity lock. Both *replay* arms win decisively (89.8% and 87.0%) because re-exposing even 15 clips per class refreshes the old boundaries *while* the new task is learned. Notably plain replay slightly beats replay+LwF here: the buffer already protects the old classes, so the extra LwF term adds a mild drag on new-class learning without a retention gain. The lesson for video is blunt—*storing a little data beats any data-free regulariser*—and it is why the capstone uses replay.

## VII. STAGE 4 — KNOWLEDGE DISTILLATION

**Why and how.** The retained model is a 25.9M-parameter ResNet50 teacher; for deployment we want something far smaller. We distil it into a MobileNetV3-Small+LSTM student ( $\sim 1.8\text{M}$  parameters,  $\sim 14\times$  smaller) and compare four objectives, all of the form  $\mathcal{L} = \alpha \mathcal{L}_{\text{CE}} + (1 - \alpha) \mathcal{L}_{\text{transfer}}$  with  $\alpha = 0.75$ :

- **CE only:**  $\mathcal{L}_{\text{transfer}} = 0$  (labels alone, the baseline).
- **KD** [6]: match softened logits,  $\mathcal{L}_{\text{transfer}} = T^2 \text{KL}(\sigma(z_t/T) \parallel \sigma(z_s/T))$ .
- **Cosine:** align the 256-d LSTM features by direction,  $\mathcal{L}_{\text{transfer}} = 1 - \cos(\varphi_s(x), \varphi_t(x))$ .
- **MSE-regressor** [7]: a learnable regressor  $g$  maps the student feature onto the teacher’s before an  $\ell_2$  match,  $\mathcal{L}_{\text{transfer}} = \|g(\varphi_s(x)) - \varphi_t(x)\|_2^2$  (the regressor is dropped at inference).

We expect the student to recover most of the teacher’s accuracy—its knowledge transfers—but the gap between objectives to be small, because the student backbone is itself pretrained and already strong.

**Result.** Table IV: the student keeps **82.7%** against the teacher’s 87.0% at  $14\times$  fewer parameters, with cosine feature distillation the best of the four objectives.

**Why.** Compression works because the teacher’s decision function transfers: a  $14\times$  smaller student lands within  $\sim 4$  points of the teacher. Among objectives, *cosine* on the LSTM state wins because it aligns the *direction* of the student’s temporal feature with the teacher’s without forcing its exact magnitude—a gentle, achievable target across two different backbones. Plain *KD* on logits helps least here: the student is

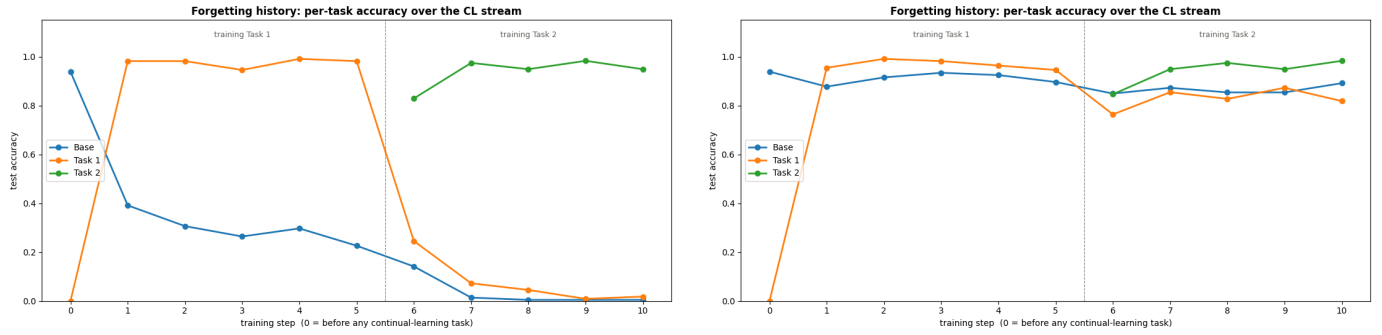


Fig. 4. Per-task accuracy across the whole stream. *Left (Naive)*: each old task’s line collapses the moment a new task is learned—catastrophic forgetting made visible. *Right (Replay)*: rehearsing a few old clips per class keeps every line high, so the final model still knows the earliest classes.

TABLE IV  
DISTILLATION INTO THE MOBILENETV3-SMALL STUDENT (20-CLASS GROUP-DISJOINT TEST). ALL STUDENTS SHARE ONE BUDGET, SEED, AND INITIALISATION; ONLY THE TRANSFER TERM DIFFERS.

| Model                          | Test acc. (%) | Params (M) |
|--------------------------------|---------------|------------|
| Teacher (Replay+LwF, ResNet50) | 87.0          | 25.9       |
| Student, CE only               | 80.6          | 1.8        |
| Student, CE + KD               | 79.7          | 1.8        |
| <b>Student, CE + cosine</b>    | <b>82.7</b>   | 1.8        |
| Student, CE + MSE (regressor)  | 75.9          | 1.8        |

pretrained, so labels already carry it far and the extra “dark knowledge” in softened logits is small. *MSE* is the harshest objective (it demands matching direction *and* magnitude between mismatched feature bases) and does worst. The honest headline of this stage is therefore *compression*—teacher-level behaviour at  $14\times$  fewer parameters—with a modest bonus from feature-cosine distillation, exactly as the ability of each objective predicts.

## VIII. STAGE 5 — FEW-SHOT META-ADAPTATION

**The question and the design.** Given the deployed 20-class student, how well can it learn *brand-new* classes from only a handful of clips? We keep the 20 base classes and add 5 new sports (a 5-way task, matching the 5-way meta-training). Every method uses the *same* few-shot adaptation—the same 8 support clips per class, the same 5 epochs, learning rate, seed, and the same replay buffer / LwF teacher objects—so only two things vary: the *initialisation* and whether replay / LwF are switched on. Three inits (*plain* student; *fine-tune*, i.e. the student after ordinary supervised training on 30 auxiliary sports; and *MAML*, the student after meta-training on the same 30 sports) are crossed with three protection settings, giving nine arms. The *fine-tune* arm is the key control: it saw the same extra data as MAML but trained in the ordinary way, so *MAML vs fine-tune* isolates meta-learning from “just more data.”

**How MAML shapes the init.** First-order MAML learns an initialisation  $\theta$  from which a few gradient steps reach a good

task-specific solution. For each episode it adapts on a support set,

$$\theta' = \theta - \alpha \nabla_{\theta} \mathcal{L}^{\text{sup}}(\theta), \quad (7)$$

then updates the initialisation by the loss of the *adapted* weights on a held-out query set,

$$\theta \leftarrow \theta - \beta \nabla_{\theta'} \mathcal{L}^{\text{qry}}(\theta'), \quad (8)$$

(first order: the gradient is taken at  $\theta'$ ). Over many episodes  $\theta$  moves to a place from which one task is a few steps away. To keep this on a free T4 and to protect the base features, only the last backbone blocks, the LSTM, and the head are meta-trained; the lower backbone stays frozen. Because meta-training shifts the features, we then re-fit a fresh linear head to the shifted features (a short linear probe) so that the MAML init is placed on equal footing with the plain init, whose head already matches its features.

**Why the metric.** With 20 base and 5 new classes, a plain overall accuracy is 20/25 base and rewards a method that guards the base and learns nothing new. We rank by the harmonic mean of Eq. (2), which weights the two sides equally and collapses to zero if either is sacrificed.

**Result.** Table V (measured; base before adapting = 0.759) ranks the nine arms. **MAML+replay is best**, reaching a balanced **0.60**, and it *beats fine-tune+replay* by 0.07 harmonic, so the gain is really from meta-learning and not merely from the extra data. Figures 5–8 show the per-method comparison, the shot-count trend, the meta-training curve, and a readable old-vs-new confusion.

**Why the winners win.** Each row follows from ability. *MAML alone* has the highest new-class accuracy (0.661) but base = 0.011: a meta-learned init is built to move its features quickly onto the new task, and with no protection it moves them all the way, erasing the old classes. *Replay alone* is the opposite (base 0.677, new 0.295): re-exposure protects the old classes but the un-primed init learns the new sports slowly. *MAML+replay* combines both abilities—a fast-adapting init *and* re-exposure—so it is best balanced (0.60). The *fine-tune* control saw the same 30 sports, which is why *fine-tune+replay* is a strong baseline (0.53); that MAML+replay still beats it by 0.07 is the clean evidence that *meta-learning itself*, not just the extra data, is responsible. The *LwF* variants over-protect: with

TABLE V

FEW-SHOT META-ADAPTATION: 20 BASE + 5 NEW (5-WAY), IDENTICAL ADAPTATION FOR EVERY ARM, RANKED BY HARMONIC MEAN. MEASURED AT 8 CLIPS/CLASS, 5 EPOCHS. “BAL.” IS THE BALANCED MEAN, “HARM.” THE HARMONIC MEAN OF EQ. (2).

| Method               | New   | Base  | Bal.  | Harm.        |
|----------------------|-------|-------|-------|--------------|
| <b>MAML+replay</b>   | 0.625 | 0.576 | 0.601 | <b>0.600</b> |
| MAML+replay+LwF      | 0.580 | 0.542 | 0.561 | 0.561        |
| fine-tune+replay     | 0.464 | 0.617 | 0.541 | 0.530        |
| fine-tune+replay+LwF | 0.446 | 0.626 | 0.536 | 0.521        |
| replay               | 0.295 | 0.677 | 0.486 | 0.411        |
| replay+LwF           | 0.223 | 0.724 | 0.474 | 0.341        |
| no-meta (plain)      | 0.527 | 0.189 | 0.358 | 0.278        |
| MAML                 | 0.661 | 0.011 | 0.336 | 0.022        |
| fine-tune            | 0.500 | 0.011 | 0.256 | 0.022        |

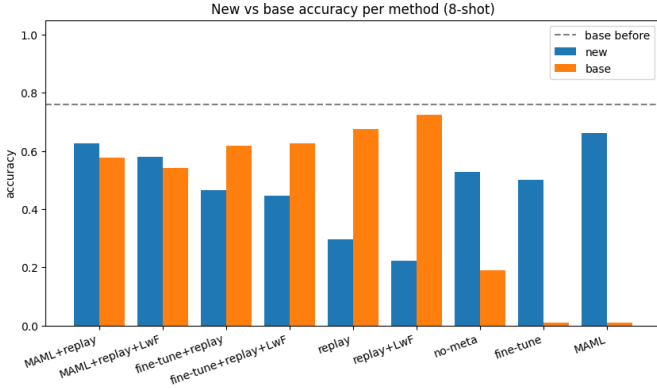


Fig. 5. New-class accuracy and base retention per method. Inits with no protection (MAML, fine-tune, no-meta) learn new classes but wipe the base; adding replay recovers the base, and the MAML init gives the highest new-class accuracy among the protected arms—so MAML+replay tops the balanced ranking.

only a few new clips, distilling the old outputs crushes the fresh new-class head, so their new accuracy is lower. Finally, the shot sweep (Fig. 6) shows the meta advantage is a property of the *init*—largest at 1 clip and shrinking as data grows—so MAML *complements* rehearsal rather than replacing it. This is also why, across the project as a whole, replay is the reliable retention tool and MAML is the tool reserved for the genuinely few-shot extension.

#### A. Cross-stage synthesis

Read together, the five stages tell one story about *regimes*. Measurement dominates everything: the group-disjoint split lowers but validates every number, and in Stage 5 the harmonic mean surfaces a MAML+replay win that a base-dominated average hides. For retention on video, *data beats cleverness*: rehearsing a few clips outperforms every data-free regulariser, because the shared weights will always drift toward the newest task unless old data pulls them back. For deployment, distillation’s win is *compression*, not a new objective—the gap between transfer losses is small once the student is pretrained. And meta-learning’s win is *few-shot extension*, largest when only one or two clips are available and vanishing as data

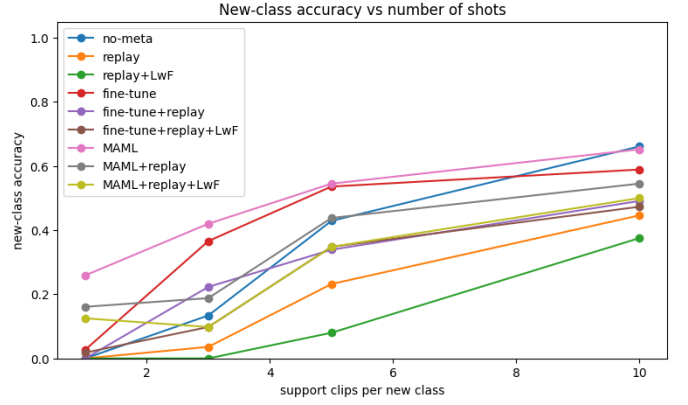


Fig. 6. New-class accuracy versus the number of support clips. MAML’s advantage is largest at **1 clip** (MAML  $\approx 0.26$  while every other init is near 0) and narrows as clips grow: meta-learning helps most exactly when data is scarce.

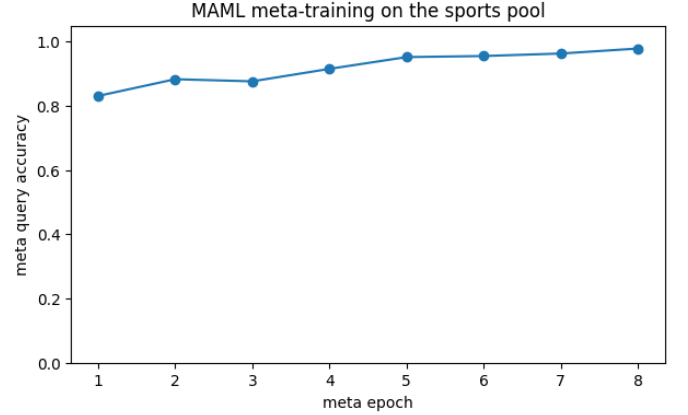


Fig. 7. MAML meta-training curve: meta query accuracy rises from 0.83 to 0.98 over 8 epochs, confirming the initialisation is genuinely being shaped rather than left at the student’s starting point.

grows. Each tool has a regime, and the pipeline is the claim that, chained in the right order, their abilities compose.

#### IX. THE ULAL CAPSTONE (TO BE RUN)

The final notebook, `ULAL.ipynb`, chains the winning choice of every stage into one lifelong pipeline and is evaluated on real video, including clips outside UCF101. *This section describes the intended run; the numbers below are expectations from the per-stage studies, not fresh measurements.*

**Stage by stage.** (1) Train the ResNet50+LSTM teacher on the base classes (the Stage-2 winner). (2) Grow it across a two-task  $10 \rightarrow 20 \rightarrow 30$  stream with *replay*, the winning retention ability from Stage 3, printing the per-task accuracy, the old-vs-new split, the confusion matrix, and the forgetting curve exactly as in the continual-learning study. (3) Clear the GPU and distil the grown teacher into the MobileNet student with the *cosine* objective (the Stage-4 winner), reporting the accuracy kept and the compression ratio. (4) Clear the GPU again and meta-train the compact student with MAML. (5) Test on real clips downloaded from the web: for *known* classes

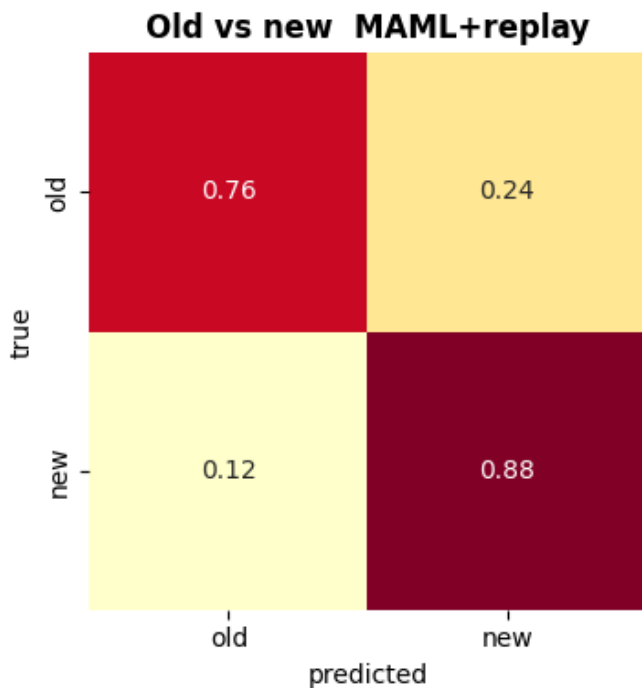


Fig. 8. Old-vs-new confusion for MAML+replay; most base clips are still predicted as base and most new clips as new. Without replay this matrix collapses, with nearly all mass predicted “new”.

this measures domain robustness (test the student directly and report accuracy and a confusion matrix); for *new* classes it measures learn-from-few (expand the head and few-shot adapt, expecting accuracy to rise from near zero after only a few examples), with a dedicated small-shot sweep over 1, 3, and 5 real clips per class.

**Why this order.** Each stage supplies an ability the next one needs—strong features, data-rich retention, deployable size, fast few-shot extension. The capstone is the concrete claim that these abilities *compose* into a single model that is accurate, small, retains old classes, and still picks up new ones from little data. Based on the per-stage results we expect the student to keep most of its known-class accuracy on real clips and to improve markedly on new classes after few-shot adaptation, mirroring the Stage-5 shot sweep. The measured numbers will be filled in once the run completes; the real-video test depends on downloadable clips, which is the capstone’s main external risk.

## X. DISCUSSION AND LIMITATIONS

Three themes recur across the pipeline. *Measurement matters*: without the group-disjoint split every later number would be inflated by scene memorisation, and without the harmonic mean the few-shot ranking would be wrong. *Data beats cleverness for retention*: on video, rehearsal of a few clips outperforms every data-free regulariser we tried (EWC, LwF), and even a small buffer makes the difference between 26% and 90% final accuracy. *Each tool has a regime*: distillation’s win is compression, and meta-learning’s win is few-shot

extension—asked outside their regime (for example MAML with plenty of data per class) they tie simpler baselines, which is itself an honest and useful finding.

**Limitations.** The study uses a single dataset and short streams (up to 20+5 classes); scaling to the full 101 classes and to longer streams is future work. Rehearsal stores real clips, so its retention is not free in memory; generative or feature-level rehearsal would remove that cost. Meta-learning’s advantage is confined to the few-shot regime, and the small student caps the achievable new-class accuracy. Finally, the capstone run and its real-video test are still pending and depend on the availability of external clips.

## XI. CONCLUSION

ULAL is a compact, honest, real-video pipeline for lifelong action recognition on a single free GPU. A leakage-free split makes the numbers trustworthy; a ResNet50+LSTM is a strong teacher (93.9%); rehearsal is the strongest continual learner (89.8% versus 26.0% for naive fine-tuning); distillation yields a 14× smaller student that keeps 82.7%; and, scored fairly with a harmonic metric and a fine-tune control, MAML+replay is the best-balanced few-shot adapter (0.60 harmonic). The organising idea—one shared recogniser with a moving freeze boundary—lets each stage change exactly one mechanism and lets their abilities compose. The capstone assembles the best of every stage into one lifelong model and is left to a final run.

## AUTHOR CONTRIBUTIONS

The work divides naturally along the pipeline of Fig. 1, and Table VI indexes which author led which part. Assignments were made so that each author owns a contiguous span of the pipeline together with a share of the shared infrastructure (the recogniser, the training harness, and the plotting and metrics code), and so that the two comparison-heavy stages (continual learning and meta-adaptation) sit with different authors. All three authors jointly designed the study, reviewed each other’s stages for fairness, and wrote the paper.

TABLE VI  
AUTHOR CONTRIBUTION INDEX, ORGANISED BY PIPELINE STAGE.

| Contributor      | Responsibilities                                                                                                                          |
|------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| Second B. Author | S1 data split and leakage audit; S2 backbone bake-off; the shared recogniser and training harness.                                        |
| Third C. Author  | S3 continual learning (all five methods, forgetting timelines); S4 knowledge distillation (four objectives, compression analysis).        |
| First A. Author  | S5 few-shot meta-adaptation (MAML, fine-tune control, harmonic metric); the ULAL capstone; the evaluation protocol, metrics, and figures. |

## REFERENCES

- [1] J. Kirkpatrick *et al.*, “Overcoming catastrophic forgetting in neural networks,” *PNAS*, 2017.

- [2] F. Zenke, B. Poole, and S. Ganguli, "Continual learning through synaptic intelligence," *ICML*, 2017.
- [3] Z. Li and D. Hoiem, "Learning without forgetting," *IEEE TPAMI*, 2017.
- [4] S.-A. Rebuffi *et al.*, "iCaRL: Incremental classifier and representation learning," *CVPR*, 2017.
- [5] M. De Lange *et al.*, "A continual learning survey: Defying forgetting in classification tasks," *IEEE TPAMI*, 2022.
- [6] G. Hinton, O. Vinyals, and J. Dean, "Distilling the knowledge in a neural network," *NIPS Deep Learning Workshop*, 2015.
- [7] A. Romero *et al.*, "FitNets: Hints for thin deep nets," *ICLR*, 2015.
- [8] C. Finn, P. Abbeel, and S. Levine, "Model-agnostic meta-learning for fast adaptation of deep networks," *ICML*, 2017.
- [9] X. Tao *et al.*, "Few-shot class-incremental learning," *CVPR*, 2020.
- [10] K. He *et al.*, "Deep residual learning for image recognition," *CVPR*, 2016.
- [11] A. Howard *et al.*, "Searching for MobileNetV3," *ICCV*, 2019.
- [12] K. Soomro, A. R. Zamir, and M. Shah, "UCF101: A dataset of 101 human action classes from videos in the wild," *arXiv:1212.0402*, 2012.